

## NELLORE, India

WMO: 432450

Lat: 14.450N

Lon: 79.983E

Elev: 20

StdP: 101.08

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 20.8       | 21.4 | 15.3                          | 10.9 | 29.5 | 16.7 | 12.0 | 28.6 | 3.5                   | 27.1 | 3.3 | 27.2 | 0.8                   | 320  | 0.335 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 9.9                    | 40.9            | 27.1 | 39.5 | 27.2 | 38.2 | 27.0 | 29.1                | 36.0 | 28.6 | 35.2 | 28.2 | 34.4 | 1.9                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.6                            | 23.6 | 31.6 | 27.1 | 23.0 | 31.4 | 26.7 | 22.4 | 31.1 | 95.1          | 36.1 | 93.0 | 35.2 | 91.1 | 34.5 | 33.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 4.3               | 3.5  | 3.2 | DB                         | 18.9 | 43.6               | 1.6 | 1.0                                                | 17.8 | 44.3       | 16.9 | 44.8       | 16.0 | 45.4       | 14.8 | 46.0 |  |
|                   |      |     | WB                         | 16.7 | 29.9               | 1.0 | 0.8                                                | 16.0 | 30.4       | 15.4 | 30.9       | 14.8 | 31.3       | 14.0 | 31.9 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 29.7   | 25.6  | 27.3  | 29.6  | 32.0  | 34.2  | 33.2  | 31.6  | 30.9  | 30.6  | 29.3  | 27.0  | 25.6 |
|                                                                               | DBStd       | 3.19   | 0.99  | 1.14  | 1.28  | 1.49  | 2.06  | 2.10  | 1.99  | 1.73  | 1.64  | 2.02  | 1.41  | 1.16 |
|                                                                               | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD10.0     | 7206   | 483   | 483   | 609   | 659   | 751   | 695   | 668   | 647   | 619   | 597   | 509   | 485  |
|                                                                               | CDD18.3     | 4164   | 225   | 250   | 351   | 409   | 492   | 445   | 410   | 389   | 369   | 339   | 259   | 227  |
|                                                                               | CDH23.3     | 52157  | 1609  | 2336  | 4143  | 5631  | 7440  | 6688  | 5854  | 5387  | 4928  | 4086  | 2422  | 1634 |
|                                                                               | CDH26.7     | 27596  | 413   | 874   | 1957  | 3258  | 4975  | 4304  | 3416  | 2961  | 2581  | 1822  | 696   | 341  |
| Wind                                                                          | WSAvg       | 1.2    | 0.8   | 1.0   | 1.2   | 1.4   | 1.4   | 1.6   | 1.6   | 1.5   | 1.2   | 0.9   | 0.9   | 1.0  |
| Precipitation                                                                 | PrecAvg     | 1046   | 7     | 2     | 5     | 12    | 22    | 46    | 86    | 89    | 125   | 277   | 306   | 78   |
|                                                                               | PrecMax     | 1742   | 38    | 13    | 86    | 100   | 124   | 143   | 175   | 310   | 361   | 758   | 772   | 329  |
|                                                                               | PrecMin     | 488    | 0     | 0     | 0     | 0     | 0     | 8     | 27    | 4     | 7     | 69    | 63    | 0    |
|                                                                               | PrecStd     | 317    | 11    | 4     | 17    | 23    | 30    | 33    | 39    | 80    | 82    | 198   | 204   | 90   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 31.2  | 34.5  | 37.7  | 40.8  | 43.1  | 41.5  | 38.7  | 37.6  | 37.6  | 36.6  | 33.3  | 31.1 |
|                                                                               |             | MCWB   | 22.9  | 23.7  | 25.4  | 26.9  | 26.9  | 27.4  | 26.5  | 26.2  | 26.0  | 25.1  | 24.8  | 23.7 |
|                                                                               | 2%          | DB     | 30.2  | 32.9  | 35.9  | 38.7  | 41.6  | 39.8  | 37.8  | 36.6  | 36.3  | 35.2  | 32.0  | 30.1 |
|                                                                               |             | MCWB   | 22.9  | 23.8  | 25.7  | 27.4  | 26.9  | 27.2  | 26.5  | 26.1  | 26.1  | 25.3  | 24.7  | 23.6 |
|                                                                               | 5%          | DB     | 29.6  | 31.9  | 34.7  | 37.4  | 40.4  | 38.6  | 36.8  | 35.8  | 35.3  | 34.2  | 30.9  | 29.4 |
|                                                                               |             | MCWB   | 22.8  | 23.7  | 25.6  | 27.2  | 27.2  | 27.0  | 26.3  | 26.0  | 26.0  | 25.6  | 24.7  | 23.4 |
|                                                                               | 10%         | DB     | 29.0  | 31.0  | 33.7  | 36.2  | 39.0  | 37.4  | 35.7  | 34.8  | 34.4  | 33.1  | 30.0  | 28.7 |
|                                                                               |             | MCWB   | 22.7  | 23.5  | 25.2  | 27.0  | 27.3  | 26.7  | 26.2  | 26.0  | 26.0  | 25.7  | 24.7  | 23.2 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 25.2  | 26.6  | 28.1  | 29.2  | 29.6  | 29.4  | 28.6  | 28.3  | 28.3  | 27.9  | 27.2  | 26.0 |
|                                                                               |             | MCDB   | 28.6  | 31.4  | 34.6  | 35.9  | 36.3  | 36.0  | 34.2  | 33.0  | 33.3  | 32.1  | 29.9  | 28.5 |
|                                                                               | 2%          | WB     | 24.6  | 25.7  | 27.4  | 28.6  | 29.1  | 28.7  | 28.0  | 27.7  | 27.7  | 27.4  | 26.6  | 25.2 |
|                                                                               |             | MCDB   | 27.9  | 30.0  | 33.1  | 35.0  | 36.2  | 36.1  | 33.8  | 32.4  | 32.4  | 31.4  | 29.3  | 27.7 |
|                                                                               | 5%          | WB     | 24.1  | 25.2  | 26.9  | 28.1  | 28.7  | 28.2  | 27.5  | 27.2  | 27.2  | 27.1  | 26.1  | 24.9 |
|                                                                               |             | MCDB   | 27.4  | 29.2  | 32.0  | 34.2  | 35.6  | 34.9  | 33.2  | 32.0  | 31.9  | 30.9  | 28.9  | 27.4 |
|                                                                               | 10%         | WB     | 23.6  | 24.7  | 26.4  | 27.7  | 28.3  | 27.7  | 27.0  | 26.9  | 27.0  | 26.7  | 25.8  | 24.4 |
|                                                                               |             | MCDB   | 27.0  | 28.7  | 31.1  | 33.4  | 34.8  | 34.2  | 32.7  | 31.8  | 31.5  | 30.4  | 28.4  | 26.9 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 7.6   | 8.6   | 9.6   | 9.8   | 9.9   | 8.3   | 7.3   | 7.2   | 7.2   | 6.5   | 5.9   | 6.3  |
|                                                                               |             | MCDBR  | 8.2   | 9.6   | 10.7  | 11.0  | 11.4  | 9.7   | 8.6   | 8.5   | 8.7   | 8.5   | 7.5   | 7.3  |
|                                                                               |             | MCWBR  | 2.7   | 2.8   | 3.0   | 2.9   | 3.2   | 3.1   | 2.8   | 2.6   | 2.7   | 2.5   | 2.4   | 2.5  |
|                                                                               | 5% WB       | MCDBR  | 6.8   | 8.3   | 9.7   | 10.2  | 10.3  | 9.0   | 8.1   | 7.8   | 7.7   | 6.9   | 5.5   | 5.4  |
|                                                                               |             | MCWBR  | 2.7   | 2.9   | 3.0   | 2.9   | 3.2   | 3.3   | 3.0   | 2.9   | 2.8   | 2.6   | 2.5   | 2.6  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.480  | 0.473 | 0.527 | 0.548 | 0.616 | 0.553 | 0.538 | 0.526 | 0.520 | 0.544 | 0.524 | 0.538 |      |
|                                                                               | taud        | 2.093  | 2.129 | 1.998 | 1.986 | 1.836 | 2.020 | 2.071 | 2.114 | 2.129 | 2.048 | 2.050 | 1.964 |      |
|                                                                               | Ebn at Noon | 812    | 840   | 803   | 783   | 719   | 757   | 771   | 790   | 797   | 768   | 766   | 744   |      |
|                                                                               | Edh at Noon | 157    | 157   | 184   | 186   | 212   | 175   | 166   | 161   | 159   | 169   | 163   | 175   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 4.86   | 5.81  | 6.35  | 6.60  | 6.13  | 5.21  | 4.81  | 4.87  | 5.00  | 4.58  | 4.25  | 4.20  |      |
|                                                                               | RadStd      | 0.18   | 0.20  | 0.19  | 0.20  | 0.21  | 0.39  | 0.45  | 0.24  | 0.32  | 0.39  | 0.38  | 0.34  |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | +0.23 | +0.46   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) |  | +0.26 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page